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# EXPERIMENT - 1 CEASER CIPHER

## AIM:-

To write a python program for implementing ceaser cipher technique.

## ALGORITHM:-

1. Start
2. Declare variables.
3. Read the text from the user.
4. Read the shift (key) values.
5. Set result as empty string to store cipher text.
6. For each character in text:
   - → If it is a letter, shift it by the key value.
   - → otherwise, keep it unchanged.
7. Display the cipher text.
8. Stop.

## CODE:-

```
text = input("Enter text:")
shift =int(input(" Enter shift value/ key value :"))
choice = input("Enter E for Encryption or D for Decryption :").upper()
result = " "
for ch in text:
        if ch.isalpha():
            if choice == "E":
                result += chr((ord(ch) -97 + shift)% 26 +97)
            elif choice == "D":
                result + = chr((order( ch) - 97 - shift)% 26+97)
            else :
                result += ch

if choice == "E":
        print("Encrypted text: ", result)
```

```
elif choice == "D":
        print("Decrypted text : ", result)

else :
        print ("Invalid choice")
```

OUTPUT :-

Enter E for Encryption or D for Decryption : E
Enter text : Hello
Enter shift value / key value : 3
Encrypted text : ptssg.

Enter E for Encryption or D for Decryption : D
Enter text : ptssg
Enter shift value / key value : 3
Decrypted text : Hello.

→ Enter text :- heel
Enter shift value / key value : 2
Enter E for Encryption or D for Decryption : E
Encrypted text : ggn.

→ Enter text : ggn
Enter shift value / key value : 2
Enter E for Encryption or D for Decryption : D
Decrypted text : heel.

RESULT :-

The program of ceaser cipher implemented successfully.

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# EXPERIMENT- 2 MONOALPHABETIC CIPHER.

## AIM:-

To write a python program for implementing monoalphabetic cipher technique.

## ALGORITHM:-

1. Start
2. Define alphabet (a-z)
3. Define key value
4. Read the plain text
5. convert text to lower text
6. Initialize result as empty string
7. For each character in the text:
   → If it is an alphabet:
   find its position in normal alphabet.
   Replace with key.
   → Else keep it unchanged.
8. Display cipher text.
9. Stop.

## CODE:-

```
plain = "abcdefghijklmnopqrstuvwxyz"
cipher = "qwertyuiopasdfghjklzxcvbnm"
choice = input ("Enter E or D : ").upper()
text = input ("Enter text : ").lower()

result = " "

if choice == "E":
        for ch in text:
            if ch in plain:
                result += index [ch]
```

```
    else:
        result += ch
    print("Encrypted text:", result)
elif choice == "D":
    for ch in text:
        if ch in cipher:
            result += plain[cipher.index(ch)]
        else:
            result += ch
        print("Decrypted text:", result)
else:
    print("Invalid choice")
```

**OUTPUT:-**

```
Enter E or D : E
Enter text : hello
Encrypted text : itssg

Enter E or D : D
Enter text : itssg
Decrypted text : hello
```

**RESULT:-**

The program of monoalphabetic cipher was implemented successfully.

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# EXPERIMENT - 3 VIGENERE CIPHER

## AIM :-

To write a python code for implementing vigenere's cipher.

## ALGORITHM :-

* Start.
* Declare variables [text, key, choice].
* Initialize result as empty string.
* Declare key-index as 0.
* For character in text :
  - → if it is an alphabet ; shift it by key value
    - ↳ if encryption :
      result += chr((ord(ch) - 97 + shift) % 26 + 97)
    - ↳ elif decryption :
      result += chr((ord(ch) - 97 - shift) % 26 + 97)
  - → else : result += ch.
* Declare key index as 1.
* Main input & show the output.

## CODE :-

```
text = input ("Enter text : ").lower()
key = input ("Enter key : ").lower()
choice = input("Enter E for encryption or D for
                decryption : ").lower()

result = " "
key_index = 0
for ch in text :
    if ch.isalpha() :
        shift = ord(key[key_index % len(key)]) - 97
        if choice == 'e' :
            result += chr((ord(ch) - 97 +
                shift) % 26 + 97)
```

```
        elif choice == 'd':
            result += chr((ord(ch) - 97 - shift) % 26 + 97)

    key_index += 1
    else :
        result += ch.
print ("Result :", result) .
```

**OUTPUT:-**

Enter text : geeksforgeeks
Enter key : ayush
Enter E for encryption or D for decryption : E
Result : gcycz fmlyleim

Enter text : gcycz fmlyleim
Enter key : ayush
Enter E for encryption or D for Decryption : D
Result : geeksforgeeks .

**RESULT:-**

The program of vigenere's cipher was implemented successfully

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# EXPERIMENT - 4 Transposition Cipher

## Aim :-

To implement a python program based on transposition method.

## Algorithm :-

Declare variables [text, key, choice]
Initialize result as empty string.
Given condition statement,

→ If of encryption:

```
for i in range (key):
    for j in range (i, len (text), key):
        result += text [j]
```

→ elif of decryption :

```
rows = len(text)
index = 0
matrix = [" "] * rows
```

And for i in range of key :

```
matrix [j] += text [index]
index += 1
```

Gain index and show the output.

Code :-

```
text = input("Enter text :").replace(" ", " ")
key = int(input("Enter number of columns : "))
choice = input("Enter E for encryption or D for
                decryption : ").lower()

result = " "
if choice == 'e':
    for i in range(key):
        for j in range(i, len(text), key):
            result += text[j]

elif choice == 'd':
    rows = len(text) // key
    index = 0
    matrix = [" "] * rows

    for i in range(key):
        for j in range(rows):
            matrix[j] += text[index]
            index += 1

result = " ".join(matrix)

print("Result :", result)
```

OUTPUT :-

```
Enter text : a good girl
Enter number of columns : 2
Enter E for encryption or D for decryption : E
Result : aodglgogr.
```

RESULT :-

Thus the transposition cipher program was successfully completed.

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# EXPERIMENT - 05

## HILL CIPHER.

**Aim :-**

To implement the python program for implementing hill cipher.

**Algorithm :-**

Declare variables & Import numpy library.

Initialize conditional statement containing if & else,

if choice is encryption :

cipher = np.dot (key, pair) % 26

elif choice is decryption :

plain = np.dot (inv_key, pair) % 26

Give input for text & choice.

Print the output as Result.

**Code :-**

```
import numpy as np
text = input ("Enter text : ").replace (" ", "").upper()
choice = input ("Enter E for Encryption or D for
                Decryption :").lower()
key = np.array ([[3,3], [2,5]])
if len (text) % 2 != 0 :
        text + = 'X'
result = " "
if choice == 'e' :
    for i in range(0, len(text), 2) :
        pair = np.array ([[ord (text [i]) - 65],
                          [ord (text [i+1]) - 65]])
```

```
        cipher = np.dot(key, pair) % 26
        result += chr(cipher[0][0] + 65) + chr(cipher[1][0] + 65)

elif choice == 'd':
        inv_key = np.array([[15, 17], [20, 9]])
        for i in range(0, len(text), 2):
                pair = np.array([[ord(text[i]) - 65],
                                 [ord(text[i+1]) - 65]])
                plain = np.dot(inv_key, pair) % 26
                result += chr(plain[0][0] + 65) + chr(plain[1][0] + 65)

print("Result:", result)
```

**Output :-**

```
Enter text : KAVIYA
Enter E for encryption or D for decryption : E
Result : EUJEUW

Enter text : EUJEUW
Enter E for encryption or D for decryption : D
Result : KAVIYA.
```

**Result :-**

Thus the Hillcipher program implemented successfully.

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# EXPERIMENT- 6 Data Encryption standard

## Aim :-

To write a python program on data encryption standard algorithm.

## Algorithm :-

Import des from cryto library.
Declare variables [choice, key]
Initialize cipher as des.new, key.encode(), des.mode_ecb
Declare condition of if & else,
If choice is 1 then
→ Give the encrypted message
→ encrypted_text = cipher.encrypt(padded_text)

elif choice is 2 then
→ Given hex value to decrypt
→ decrypted_text = unpad (cipher.decrypt (encrypted_bytes), des.block_size)

else then
→ Print Invalid choice

Gain the input & run to get the output.

## Code :-

```
from crypto.cipher import DES
from crypto.Util.Padding import pad, unpad

print("1. Encrypt")
print("2. Decrypt")
choice = input ("Enter your choice (1/2): ")
key = input("Enter 8- character key:")
```

```
if len(key) != 8:
    print("Key must be exactly 8 characters!")
    exit()

cipher = DES.new(key.encode(), DES.MODE_ECB)

if choice == "1":
    message = input("Enter message to Encrypt:")
    padded_text = pad(message.encode(), DES.block_size)
    encrypted_text = cipher.encrypt(padded_text)
    print("Encrypted text:", encrypted_text.hex())

elif choice == "2":
    encrypted_hex = input("Enter hex value to decrypt:")
    encrypted_bytes = bytes.fromhex(encrypted_hex)
    decrypted_text = unpad(cipher.decrypt(encrypted_
                        bytes), DES.block_size)
    print("Decrypted message:", decrypted_text.decod

else:
    print("Invalid choice")
```

**Output :-**

```
1. Encrypt
2. Decrypt
Enter your choice (1/2): 1
Enter 8-character key: 12345678
Enter message to encrypt: KAVIYA
Encrypted (hex): 70f181 76aa3c1a32
```

**Result :-** DES

Thus the program was intiated successfully.